

Malware (short for malicious software) refers to any program or file designed to harm, exploit, or disrupt computer systems, networks, or devices. Malware can steal sensitive information, corrupt files, disrupt operations, or give attackers unauthorized access to systems.

1. Types of Malware:

1. Viruses

- **Definition:** Malicious code that attaches itself to legitimate programs or files.
- **Impact:** Replicates and spreads when the infected program is executed, corrupting or deleting data.
- **Example:** *ILOVEYOU Virus* (2000) caused \$10 billion in damages worldwide.

2. Worms:

- **Definition:** Standalone malware that replicates itself to spread to other computers, often through networks.
- **Impact:** Consumes bandwidth, slows down networks, and causes widespread disruption.
- **Example:** *Blaster Worm* (2003) targeted Microsoft Windows, causing system crashes globally.

3. Trojans:

- **Definition:** Disguises itself as legitimate software to trick users into installing it.
- **Impact:** Provides remote access to attackers, enabling data theft, spying, or system control.
- **Example:** *Zeus Trojan* was used to steal banking information.

4. Ransomware:

- **Definition:** Encrypts the victim's data, demanding payment (ransom) for decryption.
- **Impact:** Locks out users from critical files, causing data loss and financial damage.
- **Example:** *WannaCry* Ransomware attack (2017) affected over 200,000 computers worldwide.

5. Spyware:

- **Definition:** Secretly monitors user activity, collecting sensitive data like passwords and browsing habits.
- **Impact:** Leads to identity theft, financial fraud, and invasion of privacy.
- **Example:** *Pegasus* spyware was used to spy on journalists and political figures.

6. Adware:

- **Definition:** Displays unwanted advertisements, often bundled with free software.
- **Impact:** Slows down devices, compromises privacy, and may introduce more malware.
- **Example:** *Fireball* adware infected over 250 million computers globally.

7. Rootkits:

- **Definition:** Provides attackers with administrative access to a system while hiding its presence.
- **Impact:** Difficult to detect, enabling long-term unauthorized control over a system.

- **Example:** *Sony BMG Rootkit* scandal (2005) secretly installed software on users' computers.
- 8. **Botnets:**
 - **Definition:** A network of infected devices controlled remotely by attackers.
 - **Impact:** Used for large-scale attacks like Distributed Denial of Service (DDoS), spamming, and data theft.
 - **Example:** *Mirai Botnet* (2016) caused major internet outages by targeting DNS providers.

Famous Malware Attack: The WannaCry Ransomware Attack

Overview:

- **Date:** May 2017
- **Impact:** Infected over 200,000 computers in 150+ countries
- **Target:** Organizations including the UK's NHS, FedEx, Renault, and government systems

How It Worked:

1. **Exploitation of Vulnerability:**
 - WannaCry exploited a vulnerability in Microsoft Windows called **EternalBlue**, which was originally discovered by the NSA and later leaked by a hacking group called the Shadow Brokers.
 - The exploit allowed WannaCry to spread rapidly without user interaction, making it a *worm-like* ransomware.
2. **Infection Process:**
 - Once a system was infected, WannaCry encrypted the user's files and displayed a ransom note demanding payment in Bitcoin (around \$300–\$600) for file decryption.
 - It used strong encryption algorithms, making it nearly impossible to recover files without the decryption key.
3. **Global Spread:**
 - The malware spread rapidly across networks, especially in organizations that hadn't installed Microsoft's security patch released months earlier.
 - Major industries, hospitals, banks, and government systems were affected, causing widespread disruption.
4. **Mitigation:**
 - A security researcher accidentally discovered a "*kill switch*" by registering a domain name found in WannaCry's code, which slowed its spread significantly.
 - Organizations applied security patches, and Microsoft even released updates for unsupported versions like Windows XP.

Impact:

- **Financial Losses:** Estimated damages of over **\$4 billion** globally.
- **Healthcare Crisis:** UK's National Health Service (NHS) was severely affected—canceling appointments, surgeries, and causing delays in patient care.

- **Global Wake-Up Call:** Highlighted the importance of regular software updates, patch management, and cybersecurity preparedness.